

计算机网络练习题（名词匹配专题）

I、Acronyms (10 points, 1 point for each acronym)

Instructions: Match the acronyms/terms with their correct definitions.

List:

A. TTL B. IMAP C. HOL Blocking D. OSPF E. ARP

F. UDP G. DHCP H. MTU I. ICMP J. SSL/TLS

- (1) An application layer protocol used by email clients to retrieve messages from a mail server, allowing organization in folders.
- (2) A network layer field used to prevent packets from circulating endlessly in a network loops.
- (3) A performance issue in switches or routers where a packet at the front of a queue prevents others from being forwarded.
- (4) A protocol used by network devices to send error messages and operational information (e.g., Ping).
- (5) A link-state routing protocol that uses Dijkstra's algorithm to find the best path within an AS.
- (6) The maximum size of a data packet (in bytes) that can be transmitted over a network layer.
- (7) A protocol that dynamically provides an IP address, subnet mask, and gateway to a host.
- (8) A connectionless transport protocol providing best-effort delivery without flow control.
- (9) A protocol responsible for resolving an IP address to a physical MAC address within a LAN.
- (10) Protocols used to provide encryption and security at the presentation/transport layer (often for HTTP).

I、Acronyms (5 points, 1 point for each acronym)

Consider the following acronyms: AIMD, BGP, CIDR, DHCP, DNS, FIN, HTTP, ICANN, MSS, MTU, NAT, RFC, CRC, RIP, RST, SYN, TCP, TLD, UDP.

Match the acronyms to the descriptions below, using each acronym exactly once. Supply exactly one answer for each item below. In some cases, an acronym might fit more than one question, but there is only one way to answer these so that every acronym is used exactly once.

- 1) An error-detecting code allows detection of corrupted data within the entire Ethernet frame.
- 2) The name of the message used to initiate a TCP connection.
- 3) The documents used by the IETF to describe protocol standards.
- 4) A widely used routing protocol that does not necessarily compute lowest-cost paths.

- 5) A network management protocol used on Internet Protocol (IP) networks, which dynamically assigns an IP address and other network configuration parameters to each device on the network.

I. **Acronyms** (10 points, 1 point for each acronym)

Instructions: Match the acronyms/terms with their correct definitions.

List:

A. RTT B. CDN C. CSMA/CA D. VLAN E. Socket

F. SYN G. WHOIS H. AS I. TDMA J. MIME

- (1) A distributed system of servers that delivers web content to a user based on their geographic location to reduce latency.
- (2) An identifier for a group of routers under the same administrative control (e.g., an ISP or a university) used in BGP routing.
- (3) The time it takes for a small packet to travel from sender to receiver and then back to the sender.
- (4) A channel access method used in wireless networks (like Wi-Fi) to avoid collisions rather than detecting them.
- (5) A standard that extends the format of email to support text in character sets other than ASCII, as well as attachments of audio, video, etc.
- (6) A logical interface between the application process and the transport layer protocol.
- (7) A network technology that allows a single physical switch to be partitioned into multiple logical networks.
- (8) A flag in the TCP header used to initiate a connection during the three-way handshake.
- (9) A channel access method where the channel is divided into time slots, and each node is assigned a slot to transmit.
- (10) A query protocol used to query databases that store the registered users or assignees of an Internet resource (like a domain name).

名词匹配专题答案

1. 答案:

1. B (IMAP) 2. A (TTL) 3. C (HOL) 4. I (ICMP) 5. D (OSPF)
H (MTU) 7. G (DHCP) 8. F (UDP) 9. E (ARP) 10. J (SSL/TLS)

2. 答案:

CRC (Cyclic Redundancy Check) 错误检测码——循环冗余校验。

SYN (Synchronize) TCP 连接建立时的握手报文。

RFC (Request For Comments) IETF 定义的协议标准文档。

BGP (Border Gateway Protocol) 域间路由协议，基于策略，不一定选最短路径。

DHCP (Dynamic Host Configuration Protocol) 动态分配 IP 地址。

3. 答案:

B (CDN): 内容分发网络，根据地理位置分发内容。

H (AS): 自治系统，BGP 路由的基本单位。

A (RTT): 往返时间。

C (CSMA/CA): 载波监听多点接入/避免碰撞，用于无线网。

J (MIME): 多用途互联网邮件扩展，支持非 ASCII 字符和附件。

E (Socket): 套接字，应用层和传输层的接口。

D (VLAN): 虚拟局域网。

F (SYN): 同步位，用于 TCP 握手建立连接。

I (TDMA): 时分多址。

G (WHOIS): 查询域名注册信息的协议。